

HAORAN (MATT) WAN

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SUMMARY

Data scientist with a strong track record of applying Bayesian modeling and mixed-effects analyses to solve real-world problems, delivering predictive insights for a variety of organizations. Adept at managing end-to-end research pipelines, from experimental design and data wrangling to model implementation and interpretation, and proficient in R/Stan, Python, and CI/CD workflows. Demonstrated strong leadership by supervising cross-functional teams on concurrent projects, resulting in peer-reviewed publications and conference presentations that transformed complex statistical insights into actionable strategies for diverse stakeholders.

EDUCATION

WASHINGTON UNIVERSITY IN ST. LOUIS, St. Louis, MO

Ph.D. (focus statistical and computational modeling)

Expected May 2026

Graduate Certificate in Quantitative Data Analysis

December 2023

M.A. in Psychological & Brain Sciences

December 2023

REED COLLEGE, Portland, OR

B.A. in Economics & Psychology

August 2017 – May 2021

SKILLS

Programming & Tools: R (Stan), Python, Stata, Git, GitLab, Excel

Statistical Methods: Bayesian inference, mixed-effects modeling, causal inference, longitudinal data analysis

Data Science: Predictive modeling, CI/CD, data visualization, multivariate analysis, experimental design

EXPERIENCE

CRITICAL PATH INSTITUTE, Tucson, AZ

2024

Quantitative Medicine Analyst Intern

- Developed and implemented Bayesian nonlinear mixed-effects models including generalized logistic and bounded integer approaches in brms and Stan to analyze Huntington's disease progression, identifying distinct trajectories for clinical measurements across the course of disease.
- Performed iterative model assessment and comparison (e.g., leave-one-out cross-validation) to refine disease-progression estimates, systematically examining error structures (e.g., additive vs. proportional) and integrating uncertainty propagation into final inferences for internal reporting and data-driven decision support.
- Employed robust Bayesian workflows, such as prior/posterior predictive checks and model visualization, to ensure reliable, reproducible results. Streamlined the data wrangling and modeling pipeline through GitLab CI/CD, reducing turnaround time for model updates and facilitating collaboration across multidisciplinary teams.

WASHINGTON UNIVERSITY IN ST. LOUIS, St. Louis, MO

2022 – Present

Part-time Lecturer; Graduate Teaching Assistant; Assistant to Instructor

- Delivered instructions on advanced statistical methods in R, such as hierarchical linear modeling, generalized linear regression, meta-analysis, multivariate modeling, and growth mixture modeling, equipping students with skills and multi-level data structures relevant to real-world analytics.
- Developed and graded weekly assignments and quizzes featuring robust R workflows, integrating data wrangling, model diagnostics, and visualization to reinforce best practices in statistical programming.

BEHAVIORAL ECONOMICS LABORATORY, St. Louis, MO

2021 – Present

Graduate Student Researcher

- Designed and executed rigorous experimental protocols to study intertemporal and risky decision-making, integrating controlled laboratory experiments with large-scale behavioral data collection to generate robust datasets.
- Conducted comprehensive statistical analyses in R using advanced methods, such as generalized/nonlinear mixed-effects models and Bayesian inference, to quantify decision-making processes. Streamlined and executed data wrangling and quality-control processes and established robust experimental pipelines that assessed model adequacy.
- Translated complex statistical models into actionable insights for both academic and applied settings, writing three peer-reviewed publications and regularly presenting findings at national conferences.
- Trained and supervised other researchers in experimental design, data collection, and advanced quantitative methods, ensuring high standards of quality assurance across ongoing projects.

Research Assistant

- Merged and cleaned large-scale panel datasets (e.g., CHNS, RUMiC) in Stata, applying rigorous data validation checks and creating new variables (e.g., enrollment share, wage measures) to enable robust analysis of China's secondary education system.
- Conducted advanced econometric analyses (e.g., 2SLS, instrumental variables, difference-in-differences) to estimate the causal impacts of vocational education on labor-market outcomes, addressing selection bias and endogeneity.
- Performed extensive bilingual literature reviews (English and Chinese) to contextualize empirical findings, synthesizing key themes and policy recommendations that informed both the research design and subsequent policy discussions.

SHENZHEN STOCK EXCHANGE, Shenzhen, China

2019

Fixed-Income Analyst Intern; Legal Analyst Intern

- Analyzed extensive government debt and corporate bond datasets to produce monthly and annual reports for policy evaluation, leveraging advanced Excel functionalities to streamline data processing and increase reporting efficiency.
- Conducted comparative research on capital market laws in the Greater Bay Area, contributing to refined guidelines for Main Board & SME Board operations through close collaboration with legal and policy teams.

LEARNING & ADAPTIVE BEHAVIOR LABORATORY, Portland, OR

2017 – 2021

Research Assistant

- Applied Bayesian and causal inference to small-N, high-density datasets to model risky decision-making behaviors in animals, providing insights applicable to predictive models of human behavior in economic and social contexts.
- Optimized R-based data wrangling and quality-control workflows, leveraging ggplot2 and Excel to generate reproducible results and rapid insights for single-subject repeated-measures designs.
- Translated the potential of token-economy studies for behavioral and neuroscientific applications, writing three peer-reviewed articles and presenting research at national conferences.

SELECTED PUBLICATIONS

Wan, H., Green, L., & Myerson, J. (2024). Delayed monetary losses: Do different procedures and discounting measures assess the same construct?. *Behavioural Processes*, 105101. <https://doi.org/10.1016/j.beproc.2024.105101>

Wan, H., Myerson, J., Green, L., Strube, M. J., & Hale, S. (2024). Age-related differences in delay discounting: Income matters. *Psychology & Aging*, 39(6), 632.

Wan, H. (2023). *Discounting of delayed and probabilistic outcomes across the adult lifespan*. (Master's Thesis, Washington University in St. Louis).

Wan, H., Myerson, J., & Green, L. (2023). Individual differences in degree of discounting, Do different procedures and measures assess the same construct? *Behavioural Processes*, 104864. <https://doi.org/10.1016/j.beproc.2023.104864>

Wan, H., Kirkman, C. F., Jensen, G., Hackenberg, T. D. (2021). Failure to find altruistic food sharing in rats. *Frontiers in Psychology*, 12:696025. doi: 10.3389/fpsyg.2021.696025